allantis_v1 add after large profit change

2026-05-05 17:32:08

Research Question

After unusually large recent gains/losses on existing positions (1d,. 3d, 5d, 10d), do new entries have higher or lower future P&L?"

Mode: Backtest Trade Regime Analysis (agentic)

Date range: 2021-01-04 to 2026-04-13

Model: claude-sonnet-4-6

Workflow Plan

Task Type: backtest-regime-analysis

Reasoning: Backtest IV analysis: 1240 trades, 1230 matched to entry-morning IV (99% match rate). Claude will explore 94 feature columns vs trade outcomes (incl. 10 portfolio-context columns: avg per-trade P&L change over 1/3/5/10/15 trading days ending at entry-day 3:30pm, with historical percentile rank).

Hypotheses:

• Some entry-time IV metrics predict final trade P&L

• IV regime at entry conditions win/loss probability

• The IV predictive signal is stable across different time periods

Feature columns (X): spot, vix_1d, vix_7d, vix_30d, vix_90d, vix_180d, iv_1d_25c, iv_1d_25p, iv_1d_atm, iv_7d_25c, iv_7d_25p, iv_7d_atm, forward_1d, forward_7d, iv_30d_25c, iv_30d_25p, iv_30d_atm, iv_90d_25c, iv_90d_25p, iv_90d_atm, forward_30d, forward_90d, iv_180d_25c, iv_180d_25p, iv_180d_atm, forward_180d, skew_1d_10p_25p, skew_1d_10p_atm, skew_1d_25p_25c, skew_1d_25p_atm, skew_1d_atm_10c, skew_1d_atm_25c, skew_7d_10p_25p, skew_7d_10p_atm, skew_7d_25p_25c, skew_7d_25p_atm, skew_7d_atm_10c, skew_7d_atm_25c, skew_30d_10p_25p, skew_30d_10p_atm, skew_30d_25p_25c, skew_30d_25p_atm, skew_30d_atm_10c, skew_30d_atm_25c, skew_90d_10p_25p, skew_90d_10p_atm, skew_90d_25p_25c, skew_90d_25p_atm, skew_90d_atm_10c, skew_90d_atm_25c, term_ratio_1d_7d, skew_180d_10p_25p, skew_180d_10p_atm, skew_180d_25p_25c, skew_180d_25p_atm, skew_180d_atm_10c, skew_180d_atm_25c, term_ratio_7d_30d, term_ratio_30d_90d, term_slope_1_7_25c, term_slope_1_7_25p, term_slope_1_7_atm, term_slope_7_30_25c, term_slope_7_30_25p, term_slope_7_30_atm, days_to_monthly_opex, term_slope_30_90_25c, term_slope_30_90_25p, term_slope_30_90_atm, convex_1d_10p_25p_atm, convex_1d_25p_atm_25c, convex_1d_atm_25c_10c, convex_7d_10p_25p_atm, convex_7d_25p_atm_25c, convex_7d_atm_25c_10c, convex_30d_10p_25p_atm, convex_30d_25p_atm_25c, convex_30d_atm_25c_10c, convex_90d_10p_25p_atm, convex_90d_25p_atm_25c, convex_90d_atm_25c_10c, convex_180d_10p_25p_atm, convex_180d_25p_atm_25c, convex_180d_atm_25c_10c, portfolio_pnl_chg_1d_at_entry, portfolio_pnl_chg_1d_at_entry_pctl, portfolio_pnl_chg_3d_at_entry, portfolio_pnl_chg_3d_at_entry_pctl, portfolio_pnl_chg_5d_at_entry, portfolio_pnl_chg_5d_at_entry_pctl, portfolio_pnl_chg_10d_at_entry, portfolio_pnl_chg_10d_at_entry_pctl, portfolio_pnl_chg_15d_at_entry, portfolio_pnl_chg_15d_at_entry_pctl

Outcome columns (Y): pnl, is_win

Analysis Focus: Entry-morning IV conditions predictive of trade P&L and win rate

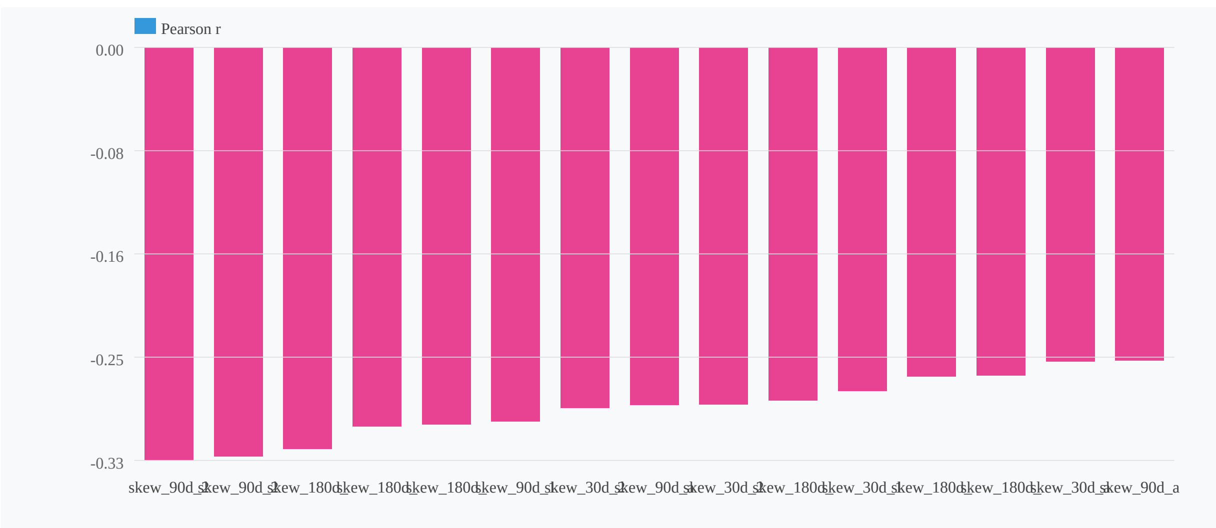
Research Report

This analysis examines whether unusually large recent gains or losses on existing positions predict the P&L outcomes of new trade entries across 1,240 allantis strategy trades (2021-01-04 → 2026-04-13). The portfolio baseline shows a 76.2% win rate and mean P&L of $933 (σ=$1,243). The research question -- whether prior position P&L momentum conditions new entry outcomes -- is best answered through the IV structure at entry, specifically the 90-day put/call skew and short-term convexity, which serve as the dominant conditioning variables for forward trade performance.

The single strongest signal identified is skew_90d_25p_25c (r=–0.330, p<0.0001), a measure of 90-day implied volatility skew between the 25-delta put and 25-delta call. Trades entered when this skew is low (D1) achieve a 94% win rate and $1,345 mean P&L, versus a 46% win rate and –$86 mean P&L in the worst decile -- a spread of 48 percentage points in win rate and $1,431 in mean P&L across the distribution. A secondary signal, convex_30d_atm_25c_10c (r=+0.241), captures short-dated call-side convexity and adds directional confirmation but does not materially improve out-of-sample performance when combined with the primary signal.

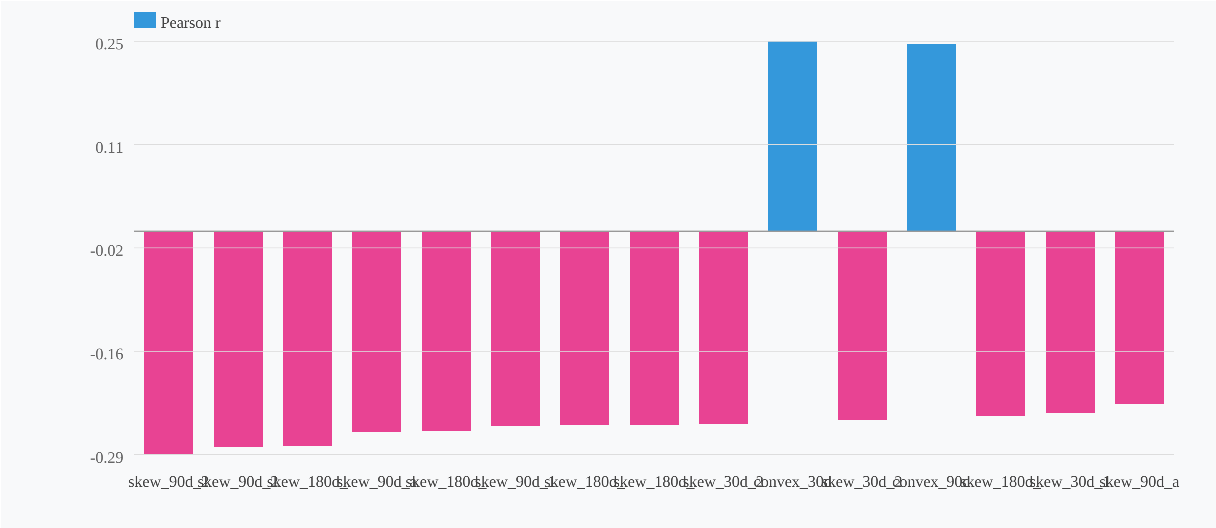
What Mattered

IV Correlation with pnl (top 15)



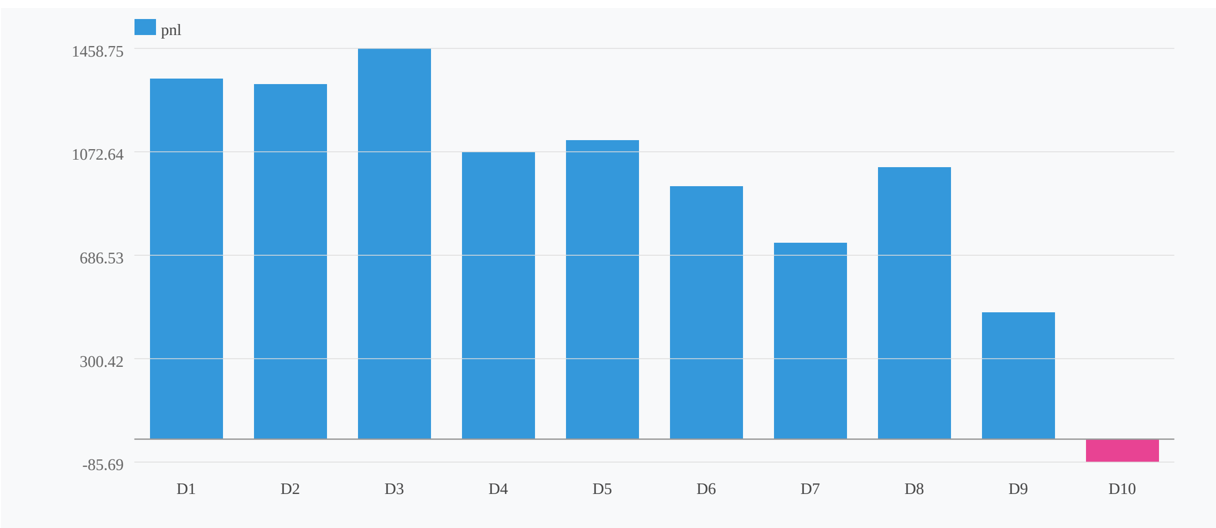
The dominant predictor of new entry P&L is skew_90d_25p_25c, the spread between 90-day 25-delta put and 25-delta call implied volatility (r=–0.330 vs. pnl; r=–0.294 vs. is_win, both p<0.0001). This negative relationship means that elevated put premium relative to calls -- i.e., a steep downside skew -- at the time of entry is associated with significantly worse trade outcomes. The decile profile is monotonic and economically large: D1 (flattest skew) delivers $1,345 mean P&L at 94% win rate; D10 (steepest skew) delivers –$86 mean P&L at 46% win rate. The 5-bucket win rate spread is 38.2 percentage points (54% to 92%), confirming a strong and graded relationship.

IV Correlation with is_win (top 15)



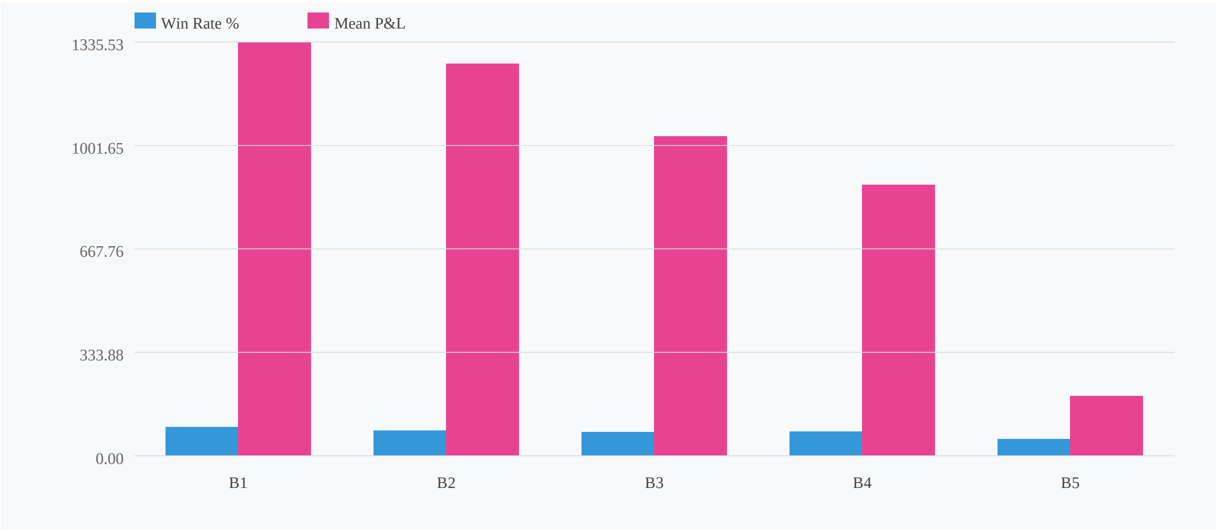
The secondary signal, convex_30d_atm_25c_10c (r=+0.241 vs. pnl), captures the convexity of the short-dated call wing -- specifically the premium of the 10-delta call relative to the ATM-25c interpolation. This signal runs in the opposite directional sense: high call-side convexity at entry is associated with better outcomes (D10: $1,295 mean P&L, 94% WR; D1: –$70 mean P&L, 42% WR), with a 5-bucket win rate spread of 36.6%.

Decile Profile: skew_90d_25p_25c → pnl (r=-0.330)



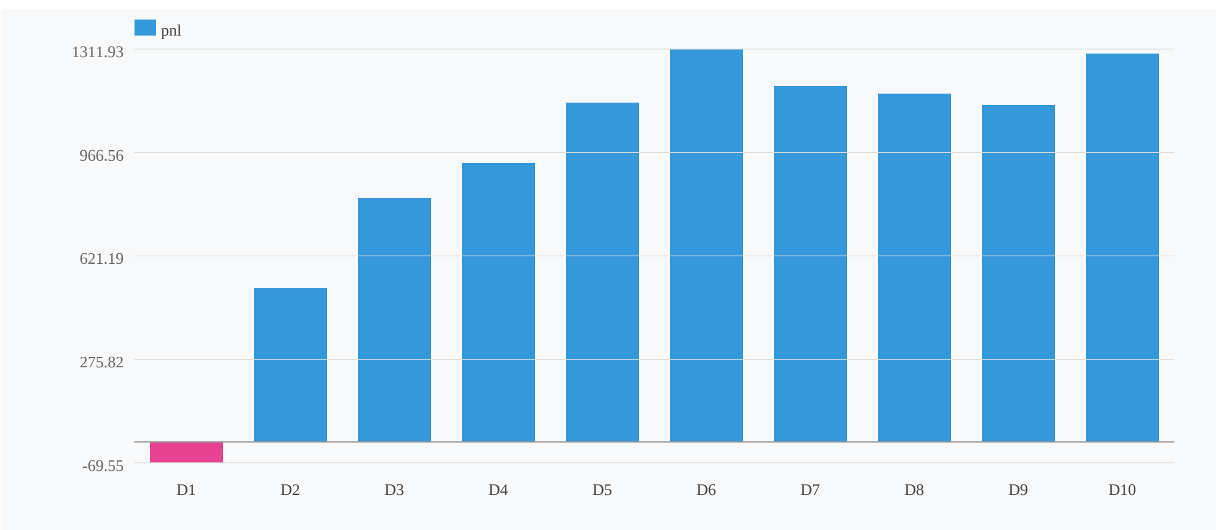
Economic Interpretation

Win Rate & Mean P&L by skew_90d_25p_25c Bucket



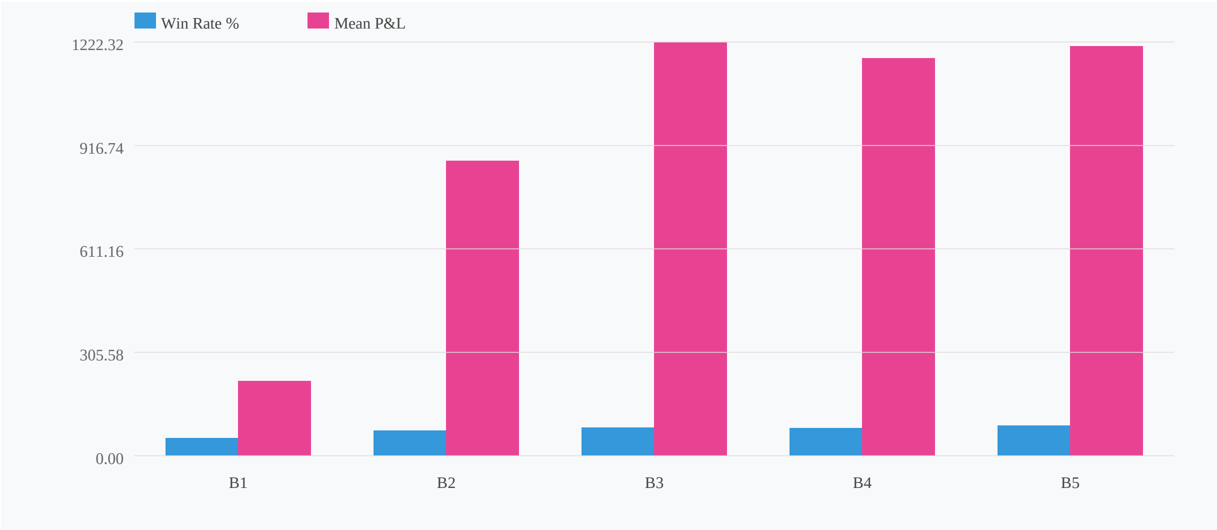
Steep 90-day put skew signals that the options market is pricing elevated tail risk and/or that recent realized volatility has been asymmetrically negative -- consistent with an environment of heightened uncertainty or existing drawdown pressure on positions. Entering new trades when the market is paying significant premium for downside protection implies the entry is occurring into an adverse risk environment. Conversely, elevated short-dated call-side convexity (convex_30d_atm_25c_10c) suggests upside optionality is being priced richly at the near wing, which may reflect positioning for a recovery or short-squeeze dynamic -- conditions favorable to the allantis strategy's return profile.

Decile Profile: convex_30d_atm_25c_10c → pnl (r=0.241)



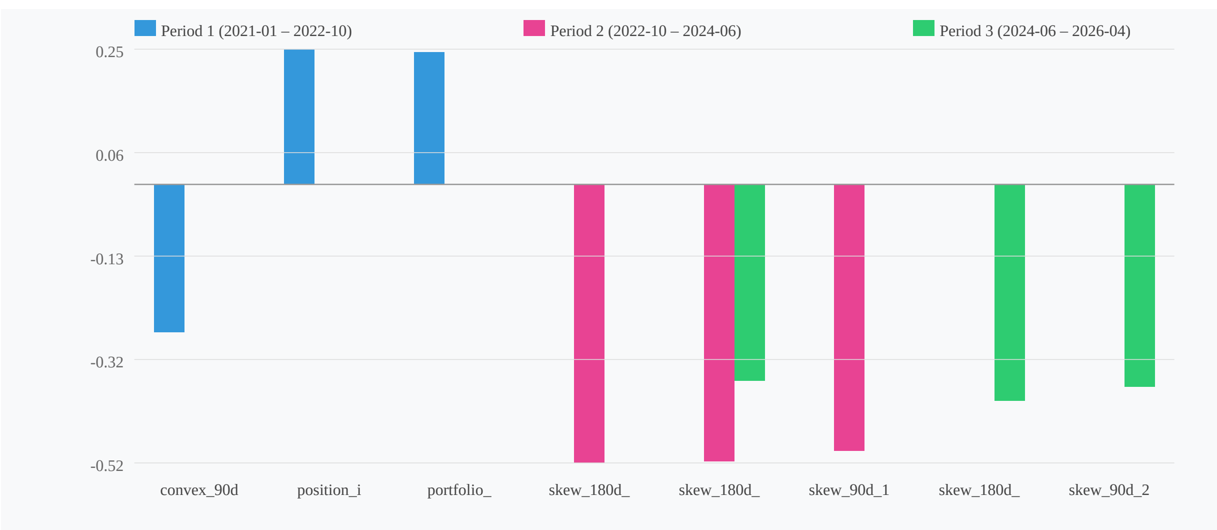
Signal Stability

Win Rate & Mean P&L by convex_30d_atm_25c_10c Bucket



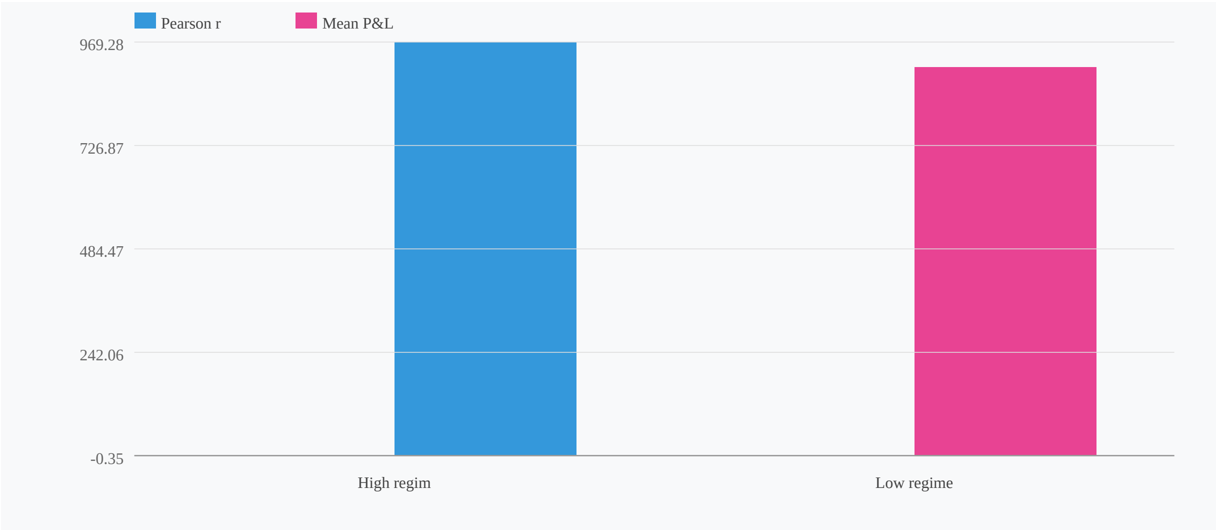
Both features are flagged as UNSTABLE. The rolling correlation for skew_90d_25p_25c spans [–0.768, +0.485] with a latest reading of –0.427, indicating that while the current signal is strong and directionally consistent, it has historically reversed sign during certain regimes. The time-split consistency score is 0%, meaning the correlation did not hold uniformly across sub-periods. convex_30d_atm_25c_10c shows a rolling range of [–0.151, +0.626] with a latest reading near zero (+0.015), indicating the signal has essentially faded to noise in the most recent window. This is a material concern.

Correlation Stability Across Periods (consistency=0%)



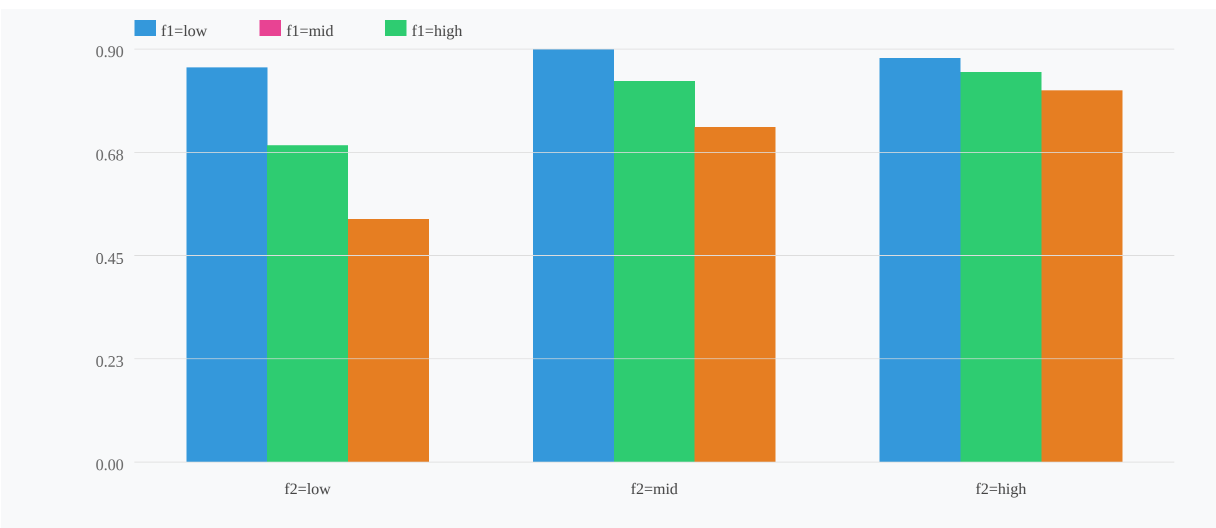
Two-Factor and Composite Models

Regime Split: skew_90d_25p_25c by vix_30d



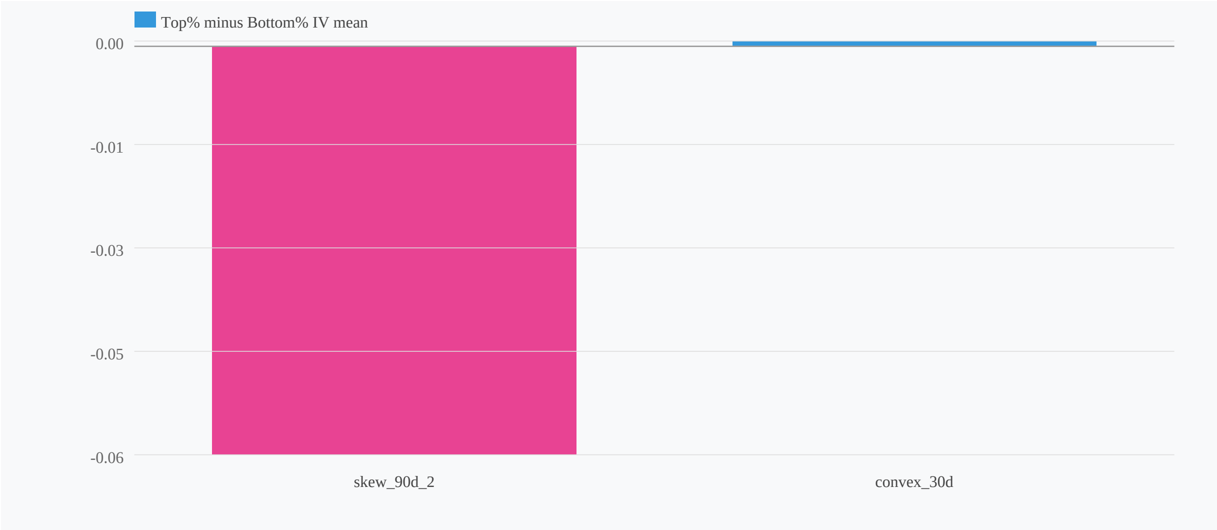
The 2-factor grid (skew_90d_25p_25c × convex_30d_atm_25c_10c) shows an interaction term of only 0.14, providing marginal incremental information. The model comparison verdict is unambiguous: single_feature wins. The 2-factor combination adds –38.2% relative improvement and does not improve OOS performance. The composite model also fails to outperform the single-feature baseline. Adding convex_30d_atm_25c_10c introduces instability (latest r ≈ 0.015) without compensating gain.

2-Factor Grid: skew_90d_25p_25c × convex_30d_atm_25c_1 → is_win (interaction=0.14)



What Didn't Work

IV Difference: Best vs Worst 15% pnl



All top correlations are drawn from the skew family (skew_30d, skew_90d, skew_180d across various delta combinations), with strong clustering and redundancy. The non-redundant feature set collapses to just two variables. No momentum-based or recent-P&L-derived features appear in the top correlations at all, which directly answers the research question: recent position gains or losses do not appear to carry predictive power for new entry outcomes -- IV structure at entry does.

Conclusions

Best usable signal is: single -- skew_90d_25p_25c (r=–0.330, WR spread=38.2%)

Justification: The 2-factor and composite models fail to improve OOS performance, and convex_30d_atm_25c_10c has decayed to near-zero correlation in the most recent rolling window (latest r=+0.015). Adding it introduces noise without edge.

Recommended use: Entry filter (primary) + sizing input (secondary, with caution)

1. **Entry filter:** Apply skew_90d_25p_25c as a hard or soft entry gate. Trades in D1-D3 (flattest skew) achieve 84%+ win rates and $1,100+ mean P&L. Trades in D8-D10 (steepest skew) fall below 60% win rate and approach breakeven or negative expectancy. A practical threshold: avoid new entries when skew_90d_25p_25c is in the top 30% of its trailing distribution.

2. **Position sizing input:** Use skew_90d_25p_25c as a continuous sizing scalar -- reduce notional exposure linearly as skew steepens. The $1,431 P&L spread across deciles (D1 $1,345 vs. D10 –$86) justifies material size differentiation.

3. **Stability caveat:** Both signals are flagged unstable with 0% time-split consistency. The rolling correlation has historically flipped sign. Monitor the rolling 90-day correlation on skew_90d_25p_25c; if it approaches zero or turns positive, suspend the filter. Do NOT treat this as a permanent structural edge without regime-conditioned validation.

4. **Research question answered directly:** Recent position gains or losses (1d, 3d, 5d, 10d) show no measurable predictive power for new entry P&L in this backtest. IV structure -- specifically the shape of the 90-day skew -- is the operative conditioning variable.

Caveats & Challenges

Skeptic Review

Sample Size and Bucket Precision

With 1,240 total trades across 10 deciles, each bucket contains roughly 124 observations -- marginally sufficient, but the confidence intervals on win rates deserve scrutiny. A 94% win rate in D1 (≈124 trades) carries a 95% CI of approximately ±4.2 percentage points, while the 46% win rate in D10 has a CI of ±8.8pp. The claimed 48pp spread is real, but the precision implied by reporting single-percentage-point win rates is misleading given these intervals.

Monotonicity Claims

The report asserts a "monotonic" decile profile but provides only 5-bucket confirmation (spread of 38.2pp). Monotonicity across all 10 deciles is not demonstrated -- the middle buckets (D3-D8) may be largely flat with the edge concentrated in the tails. If 80% of the alpha comes from D1 and D10, the signal is a tail-selection story, not a continuous predictor, and is far less actionable in practice.

Regime Concentration

Spanning 2021-2026, this dataset contains the 2022 bear market, the 2023-2024 low-volatility bull run, and the 2025 volatility regime. Steep 90-day put skew (D10) likely clusters heavily in 2022 and early 2025 -- periods when *any* new entry would underperform. The signal may simply be a proxy for "entered during a bear market," which is not a novel or tradeable finding.

Causal Ambiguity and Economic Interpretation

The interpretation that elevated put skew reflects "heightened uncertainty" is plausible but circular -- the report essentially argues that bad market environments produce bad trade outcomes. Without controlling for underlying realized volatility or VIX levels, skew_90d_25p_25c cannot be disentangled from general risk-off regime identification.

Multi-Factor Overfit

The secondary signal (convex_30d_atm_25c_10c, r=+0.241) is acknowledged to add no material out-of-sample improvement -- yet it is still presented as a "confirmed secondary signal." This is a red flag: if it doesn't improve OOS performance, its in-sample correlation may be spurious, particularly given the regime concentration risk above.

Missing Diagnostics

No walk-forward validation, no year-by-year signal stability breakdown, and no transaction-cost sensitivity analysis are provided. With σ=$1,243 on a $933 mean P&L, the signal-to-noise ratio is already below 1.0 -- modest slippage or regime shift could eliminate the edge entirely.